

SPARE

Calibration Program and Camera-Only Force Sensing System

Comprehensive Technical Development, Data, Modeling, Validation, Use, and Limitations Report

Purpose: thesis-source dossier for transfer into ChatGPT Work. This report consolidates project-owned code, data products, dated decisions, model evidence, physical validation, and claim boundaries as of 18 August 2026.

Field	Value
Document status	Technical evidence synthesis — share with explicit caveats
Evidence cutoff	18 August 2026 (Asia/Taipei)
Primary software	SPARE Camera and Load-Cell Calibration GUI, application/config schema 1.1.0
Experimental live model	live-sensor-experimental-hybrid-v4; validated claim not allowed
Authoring basis	Local workspace evidence; no external web sources used

Prepared for thesis development and evidence traceability
De La Salle University
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Technical Summary

Bottom line: The project has a physically exercised, integrity-oriented Windows acquisition and calibration program; a strong retrospective camera-only event-localization result; and a deliberately limited approximate force output. The software and synchronized physical acquisition are verified more strongly than the force-estimation claim. The v4 model is experimental, post-hoc, device-specific, and not eligible for a validated or deployment-grade force-sensor claim.

The SPARE Camera and Load-Cell Calibration GUI is a standalone PySide6 application for collecting manual fixed-label presses or Ender 3-driven repeated presses from a nine-region sensing skin. It acquires unannotated camera frames, Arduino Nano/HX711 load-cell samples, and—when selected—independent Marlin printer motion. It performs per-pixel optical baseline correction, signed load-cell calibration, host-monotonic synchronization, structured export, graph generation, partial-file recovery, and optional live camera-only inference. [E01, E13-E22]

The strongest present model result is discrete event-level localization: 93.43% macro F1 and 93.41% accuracy on 102 single-press events from 54 independent manual sessions, using six complete TEST-group outer folds. Every ROI's reported event recall is at least 90.28%. A selective output reaches 100% macro F1 at 82.93% coverage. These are retrospective post-hoc recovery results because the same six outer TEST groups had already been inspected during prior model development. [E04, E27-E30]

The current force output is a 32-knot monotonic isotonic mapping from a filtered, normalized signed optical spatial-range score to an approximate force in a narrow nominal band of 1.7-3.0 N. The conditional MAE is 0.328 N and the cross-validated 95th-percentile absolute error is 0.643 N. Those headline errors do not establish force resolution: the model improves MAE over a session-balanced constant by only 0.013 N (3.81%) in the retained v2 mapping, the expanded per-fold search improves by 4.49%, mean Spearman rank correlation is only 0.207, and predicted spread is about 20.7% of true spread. Force is withheld outside fitted score support and when multiple rods are active. [E27-E30]

The immutable feature-store reconciliation is internally strong: 245 sessions and 179,176 decoded frames were recovered exactly, with zero missing or unmatched video frames and unchanged archive hashes. The manual archive contributes 83 sessions/29,318 frames; the automatic archive contributes 162 sessions/149,858 frames. Governance is asymmetric: the manual archive controls application modeling and all general characterization metrics; the automatic archive is prohibited from application-model preprocessing, fitting, thresholds, ranges, selection, and evaluation, and is allowed only for hysteresis and creep-related exploratory characterization. [E02-E05, E23-E26]

Table 1. Present system and evidence status

Area	What is presently supported	Evidence status	Thesis-safe interpretation
Program acquisition/export	Camera, load cell, synchronized rows, optional printer state, videos, CSV/JSON/PNG	Implemented; physically exercised	A research acquisition/calibration platform
Load-cell calibration	Signed 200 g factor, tare, SNR/CV gates, $\pm 5\%$ verification	Physically verified on one fixture	Reference-force acquisition was demonstrated on the tested setup
Camera-only contact/event detection	Warm-up-calibrated event state with two-frame debounce	Post-hoc retrospective	Promising experimental detector; not prospectively confirmed
Discrete localization	Nine ROI classes; simultaneous-rod representation in runtime	Single-press post-hoc evidence	Strong discrete localization within this dataset; multi-press accuracy unknown
Force estimation	Approximate isotonic force inside optical support	Experimental; release blocked	Error-bounded approximation, not validated force measurement or resolution
Automatic repeated press	State machine, safety checks, exports, simulation and read-only printer checks	Software verified; full physical sequence pending	Automation infrastructure exists, but physical repeated-press safety/performance is incomplete

Area	What is presently supported	Evidence status	Thesis-safe interpretation
Sensor characterization	Sensitivity, nonlinearity, repeatability, cross-talk, noise/drift, SNR, hysteresis, lag proxies	Qualified by metric-specific boundaries	Characterization evidence must be reported with its individual protocol limitations

Source: Consolidated from [E01-E08, E23-E30].

Table 2. Headline quantitative evidence

Metric	Result	Population/grain	Qualification
Reconciled feature store	245 sessions; 179,176 frames	Independent session; decoded frame	0 missing and 0 unmatched video frames
Manual application corpus	83 sessions; 29,318 frames	54 primary + 11 no-contact + 18 replay	Only 54 primary sessions enter six TEST-group model folds
Event localization	Macro F1 93.43%; accuracy 93.41%	102 events; 54 independent sessions	Post-hoc grouped outer-fold evidence
Selective localization	Macro F1 100%; coverage 82.93%	Eligible single-press events	Abstention trades coverage for accuracy
Contact detection	Minimum fold recall 94.20%; maximum no-contact frame FPR 2.44%	Held-out TEST group and dedicated no-contact frames	Post-hoc; zero false-contact episodes in this replay
Approximate force	Conditional MAE 0.328 N; p95 absolute error 0.643 N	1.7-3.0 N; 4,622 frames/54 sessions in expanded benchmark	Not resolution; not untouched validation
Force tracking	Spearman rho 0.207; predicted/true SD ratio 0.207	Six outer folds	Fails resolution/tracking gate
Physical camera	14.985 frames/s over 60 s	901 DirectShow frames	Nominal 30 FPS readback was not authoritative
Physical load cell	~10.89 samples/s; 0.493% verification error	661 samples/60 s; one 200 g verification	One tested fixture and placement
Current automated tests	490 passed in isolated file groups	Current workspace on 18 Aug 2026	Monolithic Windows Qt run can trigger pyqtgraph GC access violation

Source: Recomputed/reconciled from [E05-E07, E23-E30, E35] and validation_checks.json.

1. Scope, Intended Use, and Evidence Hierarchy

1.1 Scope of this dossier

This report covers the complete project-owned calibration program and its evidence chain: intended purpose; hardware and software architecture; development chronology; operator workflow; camera, load-cell, and printer handling; raw and derived data; feature extraction; synchronization; immutable feature-store construction; characterization metrics; historical and current modeling; validation; limitations; thesis-safe claims; and reproducibility. Installed Arduino tooling, the local virtual environment, caches, and other third-party runtime files are not treated as project evidence except where a measured environment version affects reproducibility.

The report is descriptive and diagnostic. It reports what was implemented and measured, why certain model and governance decisions were made, and which claims remain unsupported. It does not infer causal material behavior from observational calibration archives and does not treat post-hoc cross-validation as prospective confirmation.

1.2 Evidence hierarchy

Table 3. Evidence hierarchy used throughout this report

Level	Evidence type	Examples	Permitted use
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Level	Evidence type	Examples	Permitted use
A	Physical verification	Camera cadence, load-cell cadence/calibration, accepted synchronized session	Support statements about the tested hardware setup and acquisition path
B	Immutable reconciled data	Archive hashes/CRC, 245-session feature store, session-indexed tables	Support corpus identity, completeness, and auditable derivation
C	Frozen/qualified characterization	Manual-first metric tables and validation report	Support metric-specific sensor observations with stated protocol boundaries
D	Post-hoc experimental model evidence	v1-v4 recovery metrics and grouped folds	Support feasibility and retrospective performance only
E	Historical/superseded analyses	Earlier automatic/manual/global linear models	Explain development evolution; not current application performance
F	Planned/unperformed gates	Known-force live test, 60-minute soak, representative usability, physical multi-press	Identify remaining work; no positive claim

Source: Evidence-policy synthesis of [E02-E08, E23-E34].

Controlling rule: When records disagree because the software evolved, the most recent dated evidence governs current state; earlier counts and schemas are retained as historical checkpoints. When a model metric conflicts with a release decision, the release decision controls the claim.

1.3 Repository condition and traceability limitation

The local Git repository has no committed history and the project directories are presently untracked. Consequently, this dossier reconstructs chronology from dated plans, decision logs, validation reports, run manifests, hashes, and artifacts—not from commit history. File modification times are not treated as authoritative development dates. This is a material traceability limitation for a thesis methods audit and should be corrected before future development by committing code, configuration, and decision records with tagged releases.

2. System Objective, Boundaries, and Terminology

2.1 Research objective

The program was built to acquire paired optical and mechanical-reference observations from a nine-location sensing skin, first to calibrate and characterize the device and then to explore whether a camera-only runtime could detect contact, localize the active region, and approximate force. The calibration GUI is therefore both an acquisition instrument and an experimental application host. Those roles must remain distinct: the load cell supplies offline reference labels and calibration evidence, whereas the deployed v4 inference path accepts camera-derived ROI features only.

2.2 System boundary

Table 4. System inputs, internal products, and outputs

Boundary element	Content	Role
Physical inputs	Arducam IMX179 frames; Nano/HX711 counts; optional Ender 3 position/commands; operator trial labels	Acquisition
Calibration state	Nine-ROI layout, unloaded per-pixel V baseline, signed counts/g factor, tare, verification record	Required context
Frame-level derived data	HSV statistics, signed/positive delta-V features, active fractions, centroids, legacy localization, synchronized force	Analysis/export
Characterization products	Sensitivity, nonlinearity, repeatability, cross-talk, noise/drift, SNR/detection, hysteresis, lag and NEF proxies	Sensor evidence

Boundary element	Content	Role
Experimental live inference	Contact/event state, one or more ROI labels, optical threshold ratio, optional approximate force	Camera-only runtime
Final artifacts	Original/auxiliary videos; CSV/JSON; PNG graphs; status/logs; model reports	Reproducibility and review

Source: [E01, E09, E13-E22].

2.3 Key terminology

- ROI means one of nine fixed rectangular regions in a 3×3 row-major layout. Localization is classification to these discrete regions, not continuous millimetre localization.
- Reference force means load-cell-derived force after signed calibration, tare, and synchronization. It is not a live input to the camera-only v4 model.
- Positive light means the sum or mean of $\max(\text{current } V - \text{baseline median } V, 0)$. Signed light retains darkening as negative values.
- Contact detection is the binary/state decision that an optical event is active. Event localization assigns the event to one or more ROI identities.
- Approximate force is the v4 isotonic output inside fitted optical-score support. It is not equivalent to validated measurement, sensor resolution, or certified force.
- Session is the independent acquisition unit. Frames and repeated cycles inside a session are repeated observations, not independent replicates.

3. Development History and Decision Record

Because the repository lacks commit history, the following chronology is reconstructed from dated project records and immutable run manifests. It explains how an acquisition program expanded into printer automation, formal characterization, and a disclosed experimental live sensor.

Table 5. Reconstructed development chronology

Date	Development or decision	Technical consequence
Before 3 Aug 2026	Core GUI, simulation, baseline, load-cell calibration, synchronization, recorder, graph export, and engineering reviews implemented	Architecture emphasized single hardware owners, bounded queues, exact schemas, partial artifacts, and fail-closed readiness
3 Aug 2026	Arducam/Nano/HX711 physical validation	DirectShow retained at measured 14.985 FPS; ~10.89 Hz load-cell stream; 200 g calibration and one accepted synchronized session demonstrated
4 Aug 2026	Ender 3 repeated-press software integration	Independent printer serial owner, motion state machine, press-zero semantics, bounds, force limit, logging, and simulation exports added
6-8 Aug 2026	Automatic and manual calibration archives collected	162 automatic sessions plus 83 manual sessions; later governed as different evidence roles
13 Aug 2026	Application modeling and characterization formally separated	Automatic archive forbidden for application model; manual-first characterization and immutable feature store frozen
13 Aug 2026	Authoritative preprocessing/range gate failed	No confirmatory candidate comparison was eligible under the original plan
13-16 Aug 2026	Disclosed recovery v1-v3	v1 restored retrospective metrics; v2 fixed runtime parity and exposed failed force-resolution behavior; v3 emphasized event signal and withheld Newton force
17 Aug 2026	Hybrid v4 assembled	v3 event detection/localization combined with v2 isotonic approximate force; expanded family comparison did not justify replacement
18 Aug 2026	Current audit	490 tests pass in isolated file groups; exact schema now contains 812 scoped data-dictionary rows; validated release remains blocked

Source: [E04-E08, E23-E30, E35].

3.1 Why the original application plan stopped

The original plan required a training-only contact threshold that simultaneously achieved at least 90% contact recall and no more than 5% no-contact false-positive rate, plus a common all-ROI force interval supported by session counts and monotonic-response rules. The frozen manual-only gate failed before confirmatory model comparison: the best development recall under the no-contact constraint was 85.16%; several required low-force bins were absent; ROI 5 had a nonpositive initial slope; and ROI 7's supported interval ended near 0.5625 N. The correct governance response was to stop the confirmatory path rather than optimize through the held-out data. [E02, E04]

3.2 Recovery models and permanent disclosure

Because additional data could not be collected, later work was explicitly labeled post-hoc recovery. v1 used a nine-frame signed-light spatial-range signal, unloaded quantiles, isotonic force, and active-fraction argmax. v2 replayed complete sequences with contact debounce, two-frame ROI switching, and a selective confidence threshold; its strong MAE coexisted with weak force variation tracking. v3 therefore made a unitless event signal the primary output and removed Newton force. v4 restored the v2 approximate force alongside the stronger v3 event/localization path, while retaining the release blockers in machine-readable form. No recovery version can become untouched confirmation evidence without a new prospectively frozen dataset. [E04, E12, E27-E30]

4. Hardware and Software Architecture

4.1 Tested hardware configuration

Table 6. Hardware components and tested identities

Component	Tested identity/configuration	Interface	Status
Camera	Arducam IMX179 Camera Module; VID:PID 1BCF:0B12	USB/OpenCV DirectShow, index 0	Physically verified
Load-cell controller	Arduino Nano / CH340; VID:PID 1A86:7523	COM3, 115200 baud	Physically verified
ADC/load cell	HX711 on Nano D4/D5; fixture-aware 200 g calibration	Fresh-conversion ASCII DATA rows	Physically verified on one fixture
Motion platform	Crealty Ender 3 V2 Neo; Marlin 1.1.6; EMERGENCY_PARSER:0	Independent COM4, 115200 baud	Read-only and small manual moves verified; repeated press not physically validated
Host	Windows 10.0.26200 / Windows 11 build string in later analysis manifests	PySide6 application	Project environment

Source: [E06, E07, E23].

4.2 Layered software design

Table 7. Project-owned software layers

Layer/path	Primary responsibilities	Important design properties
app.py	Application entry point and simulation switch	Launches one desktop process
core/	Configuration models, lifecycle/readiness, logging, model-bundle contracts, live inference	Dataclasses, explicit validity, state transitions, hash/schema checks
services/	Camera, load-cell serial, printer serial, repeated-press controller, Qt workers, simulation, recorder	One hardware owner per device; bounded queues; generation IDs reject stale callbacks
processing/	Orientation, ROI, baseline, HSV features, motion magnification, legacy localization, load-cell calibration, synchronization	Deterministic non-Qt functions shared by physical and simulation paths
data/	Canonical schemas, incremental CSV writers, synchronized finalization, graph export	Exact column order, missing-value rules, atomic publication
gui/	Nine workflow tabs, dual video, controls, plots, readiness, logs	Newest-only preview buffer is independent of analytical/recording queues

Layer/path	Primary responsibilities	Important design properties
arduino/	HX711 firmware	Reads only when ready; emits one ID/timestamp/raw value per fresh conversion
scripts/ and analysis_outputs/	Dataset construction, characterization, modeling, replay, performance and audit	Run manifests, hashes, grouped validation, immutable outputs
tests/	Hardware-free unit, integration, simulation, GUI-offscreen, contract and reliability tests	490 pass when isolated in current audit

Source: [E01, E08, E13-E22, E35].

4.3 Concurrency and ownership

Camera acquisition, optical analysis, motion magnification, serial load-cell acquisition, printer control, recording, and graph export operate through separate owners/workers. Preview uses a newest-only buffer so a slow UI does not back up the camera. Recording uses a bounded loss-detecting queue, while graph writes are asynchronous. The load-cell COM port and printer COM port are intentionally independent. Hardware callbacks carry transaction/generation identities so callbacks from a cancelled or superseded operation cannot mutate current state. [E08, E21]

4.4 Fail-closed scientific states

The live path defines STARTUP, SETUP_BLOCKED, BASELINE_CAPTURING, READY, LIVE_NO_CONTACT, LIVE_CONTACT, LIVE_UNCERTAIN, LIVE_LIMIT, PAUSED, RECONNECTING, and ERROR transitions. Exact localization is permitted only in contact state; incompatible layout, missing baseline, stale/non-finite features, out-of-support force scores, or model-bundle integrity errors withhold output. The experimental bundle loader enforces an allowlist, regular-file and size constraints, per-file SHA-256, a complete SHA256SUMS list, monotonic force knots, the exact feature order, and the explicit experimental release decision. [E18, E19, E28-E30]

5. Installation, Setup, and Operator Use Process

5.1 Software installation and launch

1. Install 64-bit Python 3.11 or later on Windows and create a virtual environment with ``py -3.11 -m venv .venv``.
2. Install dependencies with ``.\.venv\Scripts\python.exe -m pip install -r requirements.txt`` and verify them with ``python -m pip check``.
3. Launch hardware mode with ``.\.venv\Scripts\python.exe app.py``, or launch clearly labeled simulation mode with ``.\.venv\Scripts\python.exe app.py --simulation``.
4. Use the Windows helpers ``setup_windows.bat`` and ``run_windows.bat`` when a guided batch workflow is preferred.

The current GUI exposes Camera View; ROI and Baseline Settings; Image and HSV Processing; Motion Magnification; Load Cell and Calibration; Recording and Synchronization; Printer Motion; Graph and Export Settings; and System Status and Logs. The original display is an unannotated copy. ROI overlays and processed views are separate display/output copies and do not alter the authoritative original frame. [E01]

5.2 Arduino/HX711 setup

5. Install an HX711 Arduino library exposing ``begin()``, ``is_ready()``, and ``read()``; open ``arduino/hx711_nano_stream/hx711_nano_stream.ino``.

6. With USB power disconnected, wire HX711 DOUT to Nano D4, SCK to D5, module-rated VCC, and common GND. Connect the load cell to E+/E- and A+/A- according to its datasheet; wire color is not standardized.
7. Upload for Arduino Nano, select the correct processor/bootloader and COM port, then close Arduino Serial Monitor before GUI connection.
8. Select the same port at 115200 baud in the GUI. A physical connection is accepted only after finite numeric readings arrive; HELLO identity alone is insufficient.

The supplied firmware streams immediately. The canonical line is ``DATA,sample_id,arduino_micros,raw_adc``; optional commands are PING, START, STOP, and STATUS. The host also supports plain numeric and several compatibility formats. The firmware calls ``read()`` only after ``is_ready()``, preventing stale cached conversions from being re-emitted. Midstream HELLO/sample-ID/micros resets create a new device-session identity without resetting the host trial clock. [E01, E22]

5.3 Camera setup and orientation

9. Connect the Arducam, refresh camera indices, and confirm the device through preview and Windows Device Manager; an index alone is not identity evidence.
10. Use DirectShow first on the verified unit. Connect, confirm dimensions and cadence, and open the native Windows camera properties dialog on the active capture handle when exposure/focus/white balance/gain need adjustment.
11. Apply software rotation first and optional horizontal mirroring second. Any change invalidates the optical baseline and may swap the ROI frame dimensions.
12. Treat measured capture cadence as authoritative. The tested DirectShow stream was about 14.985 FPS despite a nominal 30 FPS property.

5.4 Nine-ROI definition and unloaded baseline

13. Define exactly nine non-overlapping, in-bounds rectangles in row-major order and save the layout if it will be reused.
14. Ensure the sensing skin is unloaded and camera settings are stable. Allow the normal camera warm-up.
15. Capture a baseline for the configured duration (default 3 s) with at least 20 valid frames.
16. Review and accept the baseline. The record binds camera device/backend/resolution/settings, processing settings, orientation, and ROI-layout identity.
17. Before recording, perform the unloaded drift check. The default acceptance criterion is mean absolute difference across the nine ROI mean-V values no greater than 5 V levels.

For each ROI, baseline capture converts every BGR frame to HSV and stores a per-pixel median V image plus a per-pixel mean V image and ROI-level H/S/V summaries. Any camera, orientation, processing, or ROI-layout change invalidates the record. [E09, E14]

5.5 Signed 200 g load-cell calibration

18. Collect an unloaded raw-count window and a loaded window using the configured known mass (default 200 g). Each normally requires at least 20 samples; 10 is a documented fallback only when measured cadence cannot supply 20 in the window.
19. Compute the signed factor ``(loaded mean - unloaded mean) / known mass``. The sign is retained so either electrical polarity produces a positive force when the same loading direction is applied.
20. Accept only if calibration SNR is at least 10, the loaded-window force CV is at most 2%, and the span is numerically nonzero.
21. Tare with a stable unloaded window; the default maximum raw standard deviation is 100 counts.

22. Reapply the known mass and require verification error no greater than $\pm 5\%$. Persist calibration identity, device/firmware/port provenance, windows, tare, and verification.

5.6 Manual recording workflow

23. Enter trial ID, sensing-skin ID, target ROI ground truth, optional specimen/participant ID, optional interaction/force class, press number, and notes. These are fixed trial labels, not frame-level contact labels.
24. Choose a new writable output root. The preflight checks path identity, writeability, free space, and non-overwrite behavior.
25. Confirm camera, exact ROI layout, accepted baseline, current drift, load-cell connection/calibration/verification, labels, output destination, and recorder/worker readiness.
26. Select the original video (required) and any optional overlay, processed, or motion-magnified streams. The authoritative quantitative source remains unannotated oriented pixels for the live application model.
27. Start recording, perform the single intended press, stop recording, and review the generated integrity/status summary and plots.

5.7 Repeated-press workflow

Printer setup begins with M115 identity, explicit G28 homing, M114 position readback, and bounds checks. Press Zero is stored in host memory and must never be implemented by G92. The target Z is ``press_zero_z_mm + signed_displacement_mm``; on the tested machine, negative displacement moves downward toward the specimen. A sequence proceeds through pre-roll, down motion, hold, up motion, inter-cycle dwell, repeated cycles, and post-roll. Command lifecycle, reported position, phase, cycle identity, and force-safety state are exported with frame and load-cell rows. [E01, E07]

Safety boundary: The force limit is a secondary software safeguard that can issue M410 when calibrated force exceeds a threshold. The tested Marlin firmware reports EMERGENCY_PARSER:0, and physical abort latency and a full repeated-press sequence were not validated. The program is not a certified machine-safety interlock; operator supervision, conservative motion bounds, mechanical clearance, and an accessible power-off path remain mandatory.

6. Acquisition, Recording, and Exported Data

6.1 Authoritative clocks and identities

Every accepted frame receives a permanent zero-based `capture_frame_id`, a `host_monotonic_ns` timestamp captured at acquisition, elapsed time relative to the recording start, and a human-readable wall clock. Every fresh HX711 conversion carries an Arduino sample ID, raw 32-bit micros timestamp, unwrapped micros value, host receipt monotonic timestamp, and a device-session identity. Host monotonic time is the authoritative cross-device clock because the camera and Arduino do not share a trigger. [E16, E20-E22]

6.2 Incremental recording and integrity

The recorder writes ``partial`` CSV/video artifacts incrementally and flushes by time and row count. Finalization is transactional: the recorder closes streams, validates exact schemas and row identities, synchronizes frames to load-cell samples, confirms video/frame/feature counts, writes status and metadata, and atomically publishes final names. If interrupted, partial recovery is fail-closed and reports why a session cannot be accepted. Original frames are never dropped silently; analytical failure produces a row with preserved identity/geometry, NaN derived values, and an error code. [E20, E21]

6.3 Standard session artifact set

Table 8. Session files and scientific purpose

Artifact	Content	Authority/notes
session_video.*	Unannotated oriented original frames	Only quantitative image source used by the immutable feature-store builder
auxiliary videos	Optional ROI overlay, processed view, motion-magnified stream	Visualization/review only unless an explicitly separate analysis declares otherwise
frame_features.csv	One row per accepted frame; camera/ROI/feature/localization/provenance fields	Current exact width: 366 columns
loadcell_raw.csv	One row per parsed physical sample; raw, calibrated, device-session, and motion fields	Current exact width: 46 columns
master_synchronized.csv	Frame row plus synchronized load-cell fields	Current exact width: 385 columns
data_dictionary.csv	Artifact-scoped column definitions, units, types, value roles, missing behavior	Current generated width: 812 dictionary rows across five schema scopes
session_config / camera / ROI / calibration / baseline files	Acquisition context and immutable identifiers	Required for provenance and replay
session_status and logs	Completion/abort, counts, errors, performance, identity checks	Acceptance evidence
spatial_graphs and graph companions	Heatmap, temporal line, and spatial-profile PNG/CSV/JSON bundles	Graph JSON controls metric meaning and units
printer_sequence files	Commands, responses, cycle/phase events, position, safety status	Present only for automated sequences

Source: [E01, E20, E21] and current_schema_dictionary.csv.

6.4 Current schema groups

Table 9. Current canonical schema composition

Group	Representative fields	Scientific role
Trial/session	session_id, trial_id, skin/specimen ID, target ROI, optional classes/notes	Ground truth and metadata fixed at trial grain
Frame identity	capture_frame_id, monotonic/wall clocks, dimensions/FPS/backend, driver readbacks	Raw identity and camera provenance
Motion provenance	sequence/cycle/phase/command, zero/targets/positions, force-limit flags	Automated-sequence context
Load-cell synchronization	nearest sample, interpolated raw, force, method/gap/offset/validity	Frame-aligned mechanical reference
Per-ROI optical	33 fields × 9 ROI: geometry, raw HSV, signed and positive delta, activity, centroids, normalized intensity	Raw-derived optical features
Legacy localization	dominant ROI, top ratios, weighted coordinates, confidence	Frame-level deterministic diagnostic
Validity/provenance	frame/baseline/load-cell validity, error, versions, calibration/baseline/layout IDs	Fail-closed interpretation and reproducibility

Source: [E20] and current_schema_dictionary.csv.

Earlier implementation reports cite 647, 734, or other data-dictionary row counts. Those values were correct at their dated checkpoints. The current code generates 812 artifact-scoped rows because later versions added printer-motion fields, richer optical signed features, graph-companion scopes, and artifact-specific definitions for same-named fields. The complete current dictionary is delivered beside this report rather than reproduced as an 812-row appendix.

7. Optical, Mechanical, and Synchronization Processing

7.1 Canonical image transform and ROI geometry

The authoritative live layout expects raw 640×480 input, then a 90° clockwise rotation followed by horizontal mirroring, producing a 480×640 analytical frame. Its fixed identity is `roi-9fbc67c50ee3bc7145fa`. Saved 480×640 originals are not transformed a second time. A raw 640×480 input is transformed exactly once. Any orientation/layout mismatch blocks the live bundle. [E02, E19, E23]

Table 10. Canonical live ROI rectangles on the 480×640 oriented frame

ROI	x	y	width	height	area (px ²)
1	98	161	82	89	7298
2	197	156	76	99	7524
3	312	154	58	97	5626
4	100	268	73	89	6497
5	204	264	76	96	7296
6	314	260	66	94	6204
7	104	383	73	77	5621
8	194	394	87	66	5742
9	312	379	61	77	4697

Source: [E02, E19].

7.2 Per-pixel optical baseline and feature formulas

For every frame, unannotated BGR is converted to OpenCV HSV exactly once. Within each ROI, the saved per-pixel baseline median V is subtracted in signed 16-bit arithmetic: $\text{signed_delta}(x,y,t) = V(x,y,t) - \text{median_baseline_}V(x,y)$. The positive response is $\text{positive_delta} = \max(\text{signed_delta}, 0)$. The implementation preserves both signed and positive summaries because darkening contains information even though legacy light-only features clip it. [E13, E14]

Table 11. Per-ROI optical feature families

Family	Fields/formula	Use
Raw HSV	circular mean H above saturation gate; mean S; mean/median/max/p95/std V	Camera/material state and saturation review
Signed response	sum, mean, median, MAD of current $V - \text{baseline median } V$	Primary v4 contact/force input uses signed sum
Positive response	mean, max, p95, sum, sum/area, std of $\max(\text{delta}, 0)$	Characterization and legacy localization
Active pixels	count and fraction where positive delta $\geq$ configured threshold (default 10 V)	v4 localization and independent rod gates
Spatial centroid	positive-delta weighted local and full-frame x/y	Spatial diagnostic
Normalized intensity	ROI mean positive delta divided by sum over nine ROIs	Legacy deterministic localization

Source: [E09, E13, E20].

7.3 Legacy frame-level deterministic localization

The general processing pipeline computes a legacy diagnostic localization from the nine ROI mean positive delta- V values. The ROI with the largest mean is returned only if it reaches the configured minimum (default 5 V). Confidence is dominant/total, with additional top-one/top-two ratio and intensity-weighted full-frame coordinates. Non-finite features make the result unavailable; invalid values are not silently replaced by zero. This legacy diagnostic is not the v4 event-localization model. [E13]

7.4 Load-cell equations

Let U be the unloaded raw mean, L the loaded raw mean, and M the known mass in grams. The signed factor is $c = (L - U)/M$ counts/g. With tare T and raw sample R , $\text{force_gf} = (R - T)/c$ and $\text{force_N} = \text{force_gf} \times 0.00980665$. Calibration SNR is $|L - U| / \max(\text{population_std_unloaded}, \text{population_std_loaded})$. Loaded-window CV is the population standard deviation of converted loaded grams-force divided by its absolute mean, times 100%. Verification is the absolute mean-force error divided by expected grams-force, times 100%. [E15]

7.5 Online frame/load-cell synchronization

A frame is first matched to an exact host-monotonic sample if available. Otherwise, linear interpolation is used only when two samples bracket the frame, both gaps are within the configured maximum (default 200 ms), the device-session identity is unchanged, and timestamps increase. If interpolation is unavailable, the nearest sample is used only within the same maximum gap; ties prefer the earlier physical sample. When no valid match exists, force fields are NaN and `synchronization_valid` is false, but the frame row is retained. Contact state is derived from synchronized force using the configured 0.05 N threshold, never from trial labels. [E09, E16]

7.6 Offline model-label alignment

Model fitting uses a separate session-safe alignment contract. Positive lag pairs an optical observation at time t with reference force at $t - \text{lag}$. Reference timestamps are reduced by median when duplicated, interpolation never crosses sessions, no extrapolation is allowed, and a bracketing gap above 300 ms produces a missing label. The six outer-fold lags were 260, 220, 240, 240, 240, and 240 ms; the all-development value was 240 ms. These lags are preprocessing hyperparameters fitted within training partitions, not intrinsic material response times. [E11, E17, E27]

7.7 Motion magnification

Eulerian color/intensity motion magnification is optional and off by default. It executes in an independent drop-stale worker and may feed visualization or explicitly requested auxiliary streams. For the authoritative live application model, quantitative features come from the original unannotated oriented frame; the load cell and printer are forbidden runtime inputs, and motion-magnified output is not substituted into the v4 camera feature contract. [E01, E02]

8. Data Sources, Extraction, Reconciliation, and Governance

8.1 Source archives and permitted roles

Table 12. Immutable archive inventory

Archive	Size / identity	Sessions / frames	Permitted role
Manual Calibration (2).zip	1.20 GiB; SHA-256 e039dcd49f3390d34dc979490dcf46a6ecc43d9e6a2aec08e4952acc14b5006a	83 sessions; 29,318 frames	Only application-model archive; controlling source for all general characterization metrics
Auto Calibration.zip	29.03 GiB; SHA-256 f81a41b635aa2b893e9fa1faff0bda2db41ad0717d8c134fb6d907e1471e344d	162 sessions; 149,858 frames	Characterization only: hysteresis and creep-related exploratory analysis

Source: [E03, E23-E25]. Full hashes are retained to prevent archive substitution.

8.2 Manual archive design

Table 13. Manual archive roles

Role	Sessions	Design	Use
Primary model/characterization	54	Six complete TEST sessions for each ROI 1-9	Outer-fold application modeling and general press characterization
Dedicated no-contact	11	Short unloaded sessions; all nine ROIs observed	No-contact false positives, noise, drift, threshold evidence
Replay-only	18	TEST1_v2 and other recovery/replay sessions	Runtime behavior and combined independent ROI gate; not primary fitting
Total	83	29,318 frames	Immutable manual corpus

Source: [E02-E05, E23-E26].

The confirmatory-style split is six outer folds, each holding out one complete TEST number across all nine ROI. Every preprocessing choice—including no-contact floor, scale, lag, contact threshold, support/range, selective threshold, and model hyperparameters—must be fitted within the outer training partition. Session is the independence grain. This structure prevents frame leakage across folds, but the subsequent recovery work remains post-hoc because the outer-fold results had already been inspected. [E02, E04, E27]

8.3 Automatic archive design

The automatic archive contains 18 sessions per ROI: three absolute displacements (1.75, 2.625, and 3.5 mm) × three speeds (200, 400, and 600 mm/min) × one nominal 20-cycle session and one nominal 10-cycle session. Thus each exact ROI × displacement × speed condition has two independent sessions, collected across two days. Cycles are nested repeated observations inside session and must not be counted as independent replicates. [E03, E24-E26]

Automatic-data firewall: Automatic sessions cannot supply application-model preprocessing, training labels, thresholds, support/range, model selection, or evaluation. They may support hysteresis, speed-stratified hysteresis, internal cycle QA, short fixed-displacement settling, and unloaded-recovery observations only. This firewall is enforced in configuration/tests and is a central thesis-methods requirement.

8.4 Feature-store extraction

28. Audit each ZIP archive by expected SHA-256, entry count, session count, and full member CRC.
29. For each session, read only the authoritative session_video original stream for quantitative pixels; do not read overlays, processed video, or motion-magnified video.
30. Reconcile every decoded frame to saved frame/master rows and exact session identity. Preserve reason-coded differences instead of silently dropping rows.
31. Apply the canonical orientation exactly once; saved 480×640 originals are not reoriented again.
32. Extract the fixed manual-only feature specification and retain raw/reference/corrected values, validity, phase, labels, layout, calibration, baseline, and source hashes.
33. For automatic sessions only, calculate a cycle-local force correction from the median synchronized unloaded dwell immediately before each cycle while preserving raw force.
34. Write one Parquet file per session, a session index, event log, reconciliation report, and run manifest. Do not store global fitted floors, scales, thresholds, Fdetect, Fmax, imputation, or model transforms in the immutable store.

Table 14. Feature-store reconciliation checks

Check	Result
Archive hashes before vs. after	Unchanged for manual and automatic archives
Sessions checkpointed	245 of 245
Rows/decoded frames	179,176 of 179,176
Missing video frames	0
Unmatched video frames	0
Canonical layout only	Passed
Frame differences reason-coded	Passed
Global fitted values absent	Passed
Build time/environment	82.51 s; CPython 3.12.13; NumPy 2.5.1; pandas 3.0.5; OpenCV 4.14.0.94

Source: [E23].

8.5 Data-quality findings

The authoritative feature-store and characterization products pass their own integrity/validation gates, but data quality is not equivalent to claim validity. The main risks are design limitations: only six primary sessions per ROI; exactly two independent automatic sessions per exact cyclic condition; two automatic collection days; short no-contact durations; zero labeled simultaneous-press events; narrow force recovery support; and no untouched confirmation set after model recovery. Four automatic sessions contribute one fewer usable cycle than nominal to cycle-bin summaries; this is disclosed and does not invalidate the session-level archive. [E24, E25]

9. Sensor Characterization Results

Characterization is governed metric by metric. Manual sessions control every general sensor-property claim. Automatic sessions contribute only conditional hysteresis and creep-related exploratory evidence. The validated characterization report classifies the package as ready to share with stated qualification boundaries, not as a universal device specification. [E03, E24-E26]

Table 15. Characterization metric-status inventory

Status	Metrics
Primary	Archive integrity/reconciliation; low-force and local sensitivity; nonlinearity; between-session repeatability; spatial uniformity; cross-talk; no-contact noise/drift
Conditional-qualified	Plateau observations; SNR/detection; hysteresis; hysteresis cycle/correction QA; noise-equivalent-force proxy
Exploratory-insufficient	Short fixed-displacement settling/recovery; approximate standalone system lag
Not established	Historical combined common range/monotonicity; true creep; true force resolution

Source: *metric_status.csv* [E26].

9.1 Static response, sensitivity, and nonlinearity

Manual low-force sensitivity is strongly position-dependent

Session-balanced slope of positive integrated light versus force over 0-1 N; zero indicates no positive fitted slope.

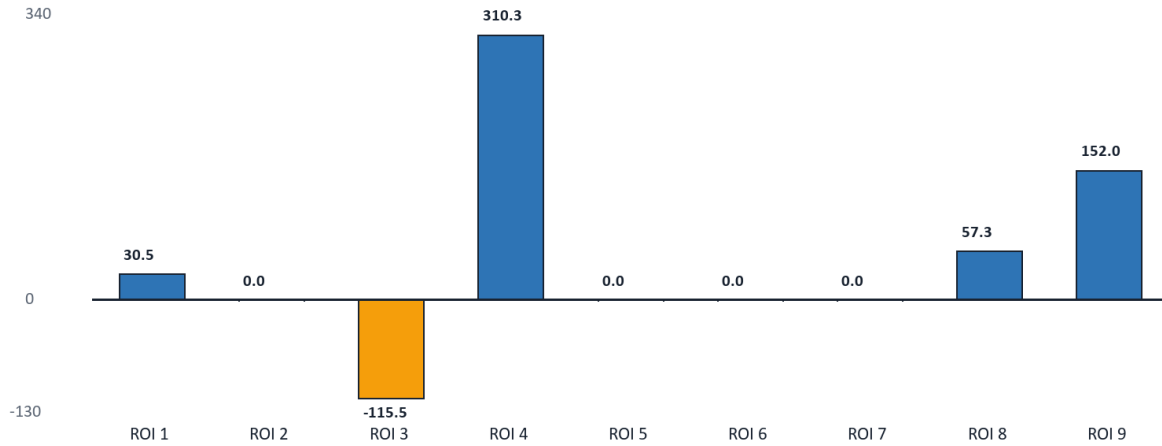


Figure 1. Manual session-balanced low-force sensitivity by ROI

Source: manual_sensitivity_nonlinearity.csv [E26]. Values are characterization slopes, not the v4 force-model coefficient.

Table 16. Manual low-force sensitivity and observed nonlinearity

ROI	Integrated light/N	Light/pixel/N	Nonlinearity	Slope interpretation
1	30.475	0.00418	—	positive
2	0.000	0.00000	—	zero/negative
3	-115.464	-0.02052	—	zero/negative
4	310.256	0.04775	—	positive
5	0.000	0.00000	—	zero/negative
6	0.000	0.00000	—	zero/negative
7	0.000	0.00000	—	zero/negative
8	57.297	0.00998	—	positive
9	151.976	0.03236	—	positive

Source: manual_sensitivity_nonlinearity.csv [E26]. Nonlinearity is reported only where the fitted slope supports the calculation.

Only ROIs 1, 4, 8, and 9 show positive low-force slopes under the frozen rule; ROI 3 is negative and ROIs 2, 5, 6, and 7 are zero after the positive-slope qualification logic. Where nonlinearity is calculable, maximum deviation spans roughly 37.7-70.1% of the observed light range. This strong spatial heterogeneity blocks a single uniform optical-force calibration and helps explain why global force models perform weakly. [E26]

9.2 Between-session repeatability and plateau observations

Manual repeatability is summarized at ROI $\times$ 0.25 N force-bin grain using independent sessions. Across 393 supported bin summaries, the median between-session standard deviation is 11.92% of observed light span and the 95th percentile is 28.93%. This is a descriptive stability measure, not an inference interval. The frozen sustained-plateau rule observed no plateau in any ROI. The reported supported-force extents are observation ranges only, not a common deployment range. [E26]

9.3 Spatial cross-talk

Single-ROI presses distribute optical response across the array

Median share of total positive response; row = labeled target, column = measured ROI. Medians are session-balanced.

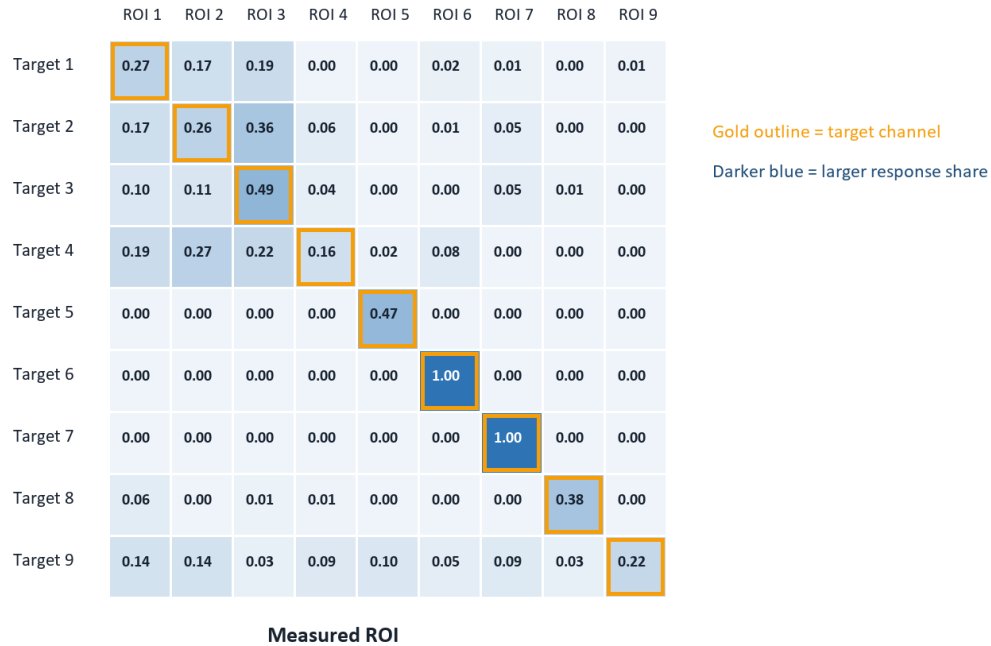


Figure 2. Session-balanced single-ROI press response distribution

Source: *cross_talk.csv* [E26]. Cell medians are calculated independently and therefore a row of medians need not sum exactly to one.

The median target-channel share across the nine diagonal cells is 0.376; diagonal values range from 0.164 to 1.000. The largest off-target median share is 0.362. Several targets distribute response across neighboring or non-target ROIs, so localization is not simply equivalent to the largest raw positive-light channel. The strong v3/v4 event result depends on normalized activity, temporal accumulation, and the labeled corpus rather than perfect optical isolation.

9.4 No-contact noise and drift

Dedicated no-contact recordings show substantial noise and short-term drift

Eleven independent no-contact sessions; signed-light MAD and median absolute drift are reported separately.

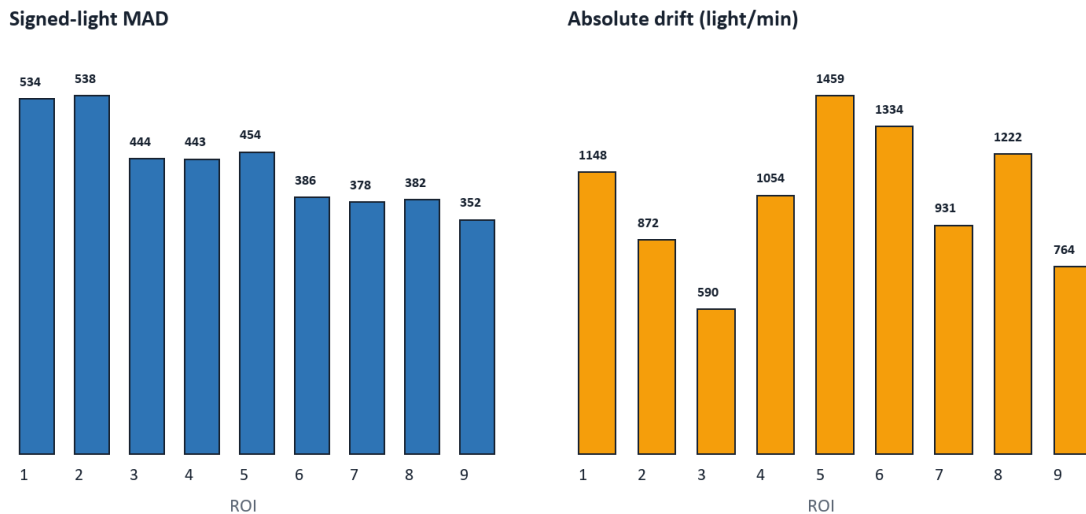


Figure 3. Dedicated no-contact signed-light noise and short-term drift

Source: *no_contact_summary.csv* [E26]; 11 independent dedicated no-contact sessions.

Table 17. No-contact stability by ROI

ROI	Signed-light MAD	Median drift /min	Maximum drift /min
1	533.5	1148.4	3340.6
2	538.5	872.3	3595.5
3	443.5	590.3	1366.9
4	443.0	1054.0	2539.0
5	454.0	1459.4	4184.5
6	386.0	1333.9	1978.7
7	378.5	931.1	2262.4
8	382.5	1222.2	3129.4
9	352.0	764.1	1872.1

Source: *no_contact_summary.csv* [E26].

Signed-light MAD ranges from 352.0 to 538.5. Median absolute drift ranges from 590.3 to 1,459.4 light units/min, and the largest observed session-level drift is about 4,184.5 light units/min at ROI 5. These are short-term recorded-session properties. They motivate the mandatory unloaded baseline, drift gate, live warm-up quantile, per-ROI normalization, and out-of-support withholding.

9.5 SNR, detection threshold, and noise-equivalent-force proxy

Table 18. Manual optical SNR in the 0.75-1.25 N band

ROI	Linear SNR	SNR (dB)
1	0.000	nan dB
2	0.000	nan dB
3	0.466	-6.64 dB
4	0.554	-5.13 dB
5	0.000	nan dB
6	0.000	nan dB

ROI	Linear SNR	SNR (dB)
7	0.000	nan dB
8	1.620	4.19 dB
9	1.213	1.68 dB

Source: *manual_snr.csv* [E26]. Zero indicates no qualified positive signal above the defined noise comparison.

Only ROI 3 at approximately 1.875 N and ROI 9 at approximately 0.875 N satisfy the frozen characterization detection rule of 1% empirical no-contact FPR and 95% recall with at least three press sessions. These are characterization observations, not application thresholds. The NEF calculation is also deliberately limited: ROIs 1, 4, 8, and 9 yield one-noise proxy values of 17.51, 1.43, 6.68, and 2.32 N respectively; other ROIs are unavailable because sensitivity is zero/negative. The proxy is noise divided by low-force slope and cannot establish true force resolution. [E25, E26]

9.6 Automatic hysteresis and cycle evidence

Automatic cyclic evidence indicates ROI-dependent hysteresis

Median absolute loading-unloading separation across automatic speed/displacement/force-bin summaries; two sessions per exact conditor

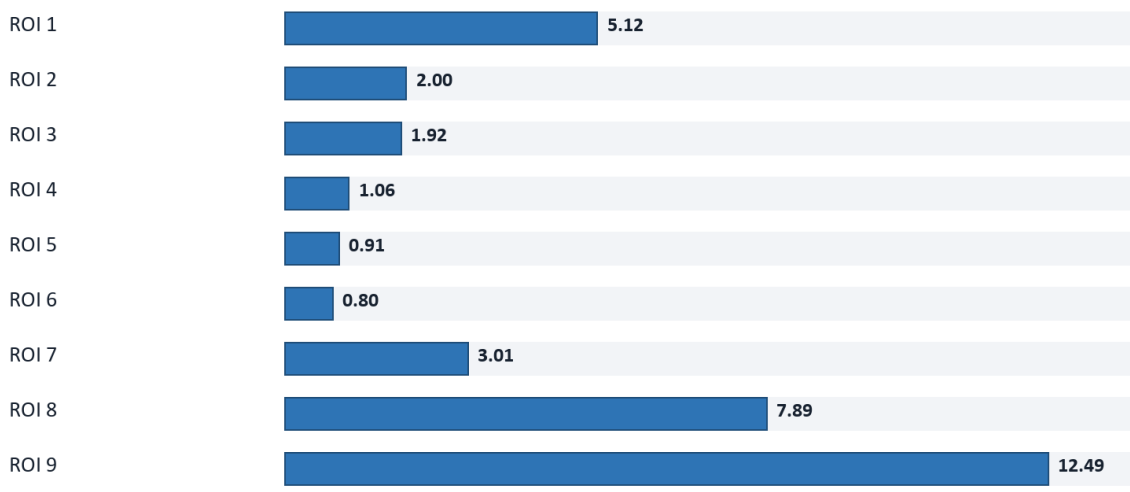


Figure 4. Median absolute hysteresis percentage by ROI

Source: *automatic_hysteresis.csv* [E26]. Two independent sessions per exact ROI × speed × displacement condition; estimates are descriptive.

Across 339 condition/force-bin summaries, the median signed hysteresis is -1.02% of observed span, the median absolute value is 2.79%, and the 95th percentile absolute value is 20.78%. The observed signed range is -37.75% to 66.83%. ROI-level median absolute hysteresis ranges from about 0.80% (ROI 6) to 12.49% (ROI 9). Speed is a hysteresis stratum, not a standalone application-model feature.

The automatic hold phase is short: median session-level cycle duration is 0.432 s. The inter-cycle dwell median is 1.936 s. These durations support short fixed-displacement settling and unloaded-recovery observations only. They do not satisfy a true creep protocol, which would require a constant-force hold of at least 5 s and preferably 30-60 s. Four sessions have one fewer usable cycle in cycle-bin summaries; the validation report identifies them explicitly. [E25, E26]

9.7 Approximate system lag

Table 19. Manual-only derivative-correlation system-lag summary

ROI	Median lag (ms)	Minimum (ms)	Maximum (ms)
1	87.5	-175.0	175.0
2	125.0	-400.0	375.0
3	137.5	-275.0	500.0
4	12.5	-325.0	200.0
5	125.0	125.0	150.0
6	25.0	-250.0	375.0
7	-12.5	-275.0	275.0
8	100.0	-350.0	500.0
9	-37.5	-275.0	150.0

Source: manual_system_lag_summary.csv [E26]. Positive means optical follows reference under the stated convention.

Median lags vary from –37.5 to 137.5 ms and several session extrema reach the search boundaries. The estimate combines camera exposure/readout, USB, host acquisition, software timestamping, reference interpolation, material response, and the selected derivative-correlation method. It is therefore an exploratory system-level lag, not intrinsic material response time or measured end-to-end display latency.

10. Force Localization Model

10.1 Current v4 event/localization pipeline

35. Read only the nine `signed\_delta\_v\_sum` and nine `active\_fraction` camera features in the exact frozen order. Load cell and printer are rejected as runtime inputs.
36. Normalize each signed sum and active fraction using manual-only per-ROI centers/scales saved in the reviewed bundle.
37. Apply a causal nine-frame moving mean to signed and active channels.
38. Calculate the contact score as the spatial peak-to-peak range of filtered normalized signed sums.
39. During an unloaded warm-up of at least 120 frames, set the global contact threshold to the larger of the bundle fallback (0.596388) and the empirical 99th percentile of warm-up scores.
40. Set nine independent rod activation thresholds to the larger of 4.0 normalized units and the warm-up 99.9th percentile plus 1.0; release occurs below 60% of the activation threshold.
41. Declare raw contact if either the global signed-range score crosses threshold or any independent active-fraction ratio reaches one. Apply two-frame acquire and clear debounce.
42. For an event, accumulate positive normalized active-fraction evidence over raw-contact frames, including the acquisition buffer. Forced ROI is the accumulated argmax; confidence is `(top1 – top2)/total`.
43. Withhold the selective single-ROI label when confidence is below 0.374959 unless independent rod gates provide active ROI identities. Require two frames before switching a maintained single-ROI candidate.
44. Represent simultaneous rods as a set when multiple independent gates are active. In that state, force is withheld because the single-press force model cannot decompose contributions.

10.2 Validation design

The reported event metrics aggregate frame evidence to events before classification. Each of six outer folds holds out the complete TEST number across all nine ROI, preserving session independence and target coverage. Model comparisons and thresholds are fitted within the outer training portion. However, the outer groups were already inspected during prior recovery, so the design is grouped retrospective evaluation rather than untouched confirmation. [E02, E04, E27]

Event-localization model comparison

Macro F1 from grouped outer-fold recovery evaluation; the simple accumulated argmax was retained.



Figure 5. Grouped event-localization model comparison

Source: v4 metrics.json [E27]. All families use 102 events/54 independent sessions; outer TEST groups had been inspected post-hoc.

Accumulated argmax was retained because it had the highest aggregate macro F1 (93.43%) compared with Extra Trees (91.14%), shrinkage LDA (89.25%), and logistic regression (85.27%). Its simplicity also matches the event physics and avoids adding a more fragile learned classifier without performance benefit.

Event-level localization recall by ROI

Six outer TEST-group folds; 102 single-press events from 54 independent sessions. Post-hoc experimental evidence.



Figure 6. Event-localization recall across all nine ROI

Source: v4 metrics.json [E27]. The 80% line is the frozen primary benchmark; post-hoc status remains controlling.

Table 20. v4 event detection and localization metrics

Metric	Value	Interpretation
Events / independent sessions / folds	102 / 54 / 6	Single-press manual TEST1-TEST6 events
Forced event accuracy	93.41%	A ROI is always selected for eligible events
Forced event macro F1	93.43%	Primary localization metric
Per-ROI event recall	90.28-100%	All nine exceed the 80% recall benchmark used for presentation; minimum is ROI 4
Selective macro F1 / coverage	100% / 82.93%	Higher precision by withholding uncertain ROI labels
Mean / minimum contact recall	97.79% / 94.20%	Across outer folds
Mean / maximum no-contact frame FPR	0.41% / 2.44%	Dedicated no-contact frames
Maximum false-contact episodes/min	0	No debounced false episode in the evaluated replay
Median detection delay	1 frame in every fold	Approximately 67 ms at 15 FPS, excluding unmeasured exposure/display latency

Source: v4 metrics.json [E27].

10.3 Simultaneous-rod handling

The runtime can display a set of independently thresholded rods and withhold force in a multi-press state. A combined replay of independent single-ROI evidence reported 100% exact-set accuracy at 83.08% displayed coverage. This is a synthetic combination/fallback check, not validation on true simultaneous physical presses: the archive contains zero labeled multi-press events. Consequently, the thesis may describe multi-ROI representability and logic coverage, but not measured multi-contact localization accuracy. [E05, E18]

11. Force Estimation Model

Required thesis wording: The model produces an approximate, post-hoc, camera-only force estimate inside a narrow optical-score support. Its cross-validated error is reportable, but force resolution, general accuracy, common safe range, and physical deployment validity are not established.

11.1 Input, mapping, and runtime behavior

The v4 force component retains the v2 monotonic isotonic mapping. Its scalar input is the filtered normalized signed spatial range: after per-ROI centering/scaling and the causal nine-frame mean, the score is $\text{max}(\text{filtered_signed}) - \text{min}(\text{filtered_signed})$. The model contains 32 strictly ordered x knots and nondecreasing y knots. Linear interpolation between knots produces approximately 1.859-2.693 N; declared force-operation labels are 1.7-3.0 N. A score below the first knot returns `below_range`, above the last knot returns `above_range`, and neither produces a numeric force. During an event, the completed-event force is the maximum in-support frame estimate. Force is set to 0 only in the hybrid NO_CONTACT state. [E18, E29]

The runtime exposes current approximate force while a single-rod event is active and peak approximate force at completion. It also records the approximate force associated with the event peak raw V and peak positive delta-V for the selected ROI. If multiple rods are active, all force outputs are withheld because the training and mapping are single-press. If the optical score leaves support, force is withheld rather than clipped to an endpoint. [E18, E29]

11.2 Candidate model search

The original plan listed zero-intercept total light, nonnegative least squares, positive ridge, elastic net, isotonic, monotonic piecewise/spline, histogram gradient boosting, XGBoost, and exploratory SVR/tree/MLP candidates.

The post-hoc expanded benchmark compared grouped isotonic, ridge, elastic net, Extra Trees, and histogram gradient boosting over 30 normalized signed/active summary features and causal windows. Outer-fold selection chose Extra Trees in four folds, isotonic in one, and ridge in one. The family instability and negligible aggregate difference from v2 did not justify replacing the reviewed monotonic mapping. [E02, E04, E27]

Force approximation versus a session-balanced constant

Outer-fold conditional MAE; lower is better. Paired within each held-out TEST group (post-hoc).

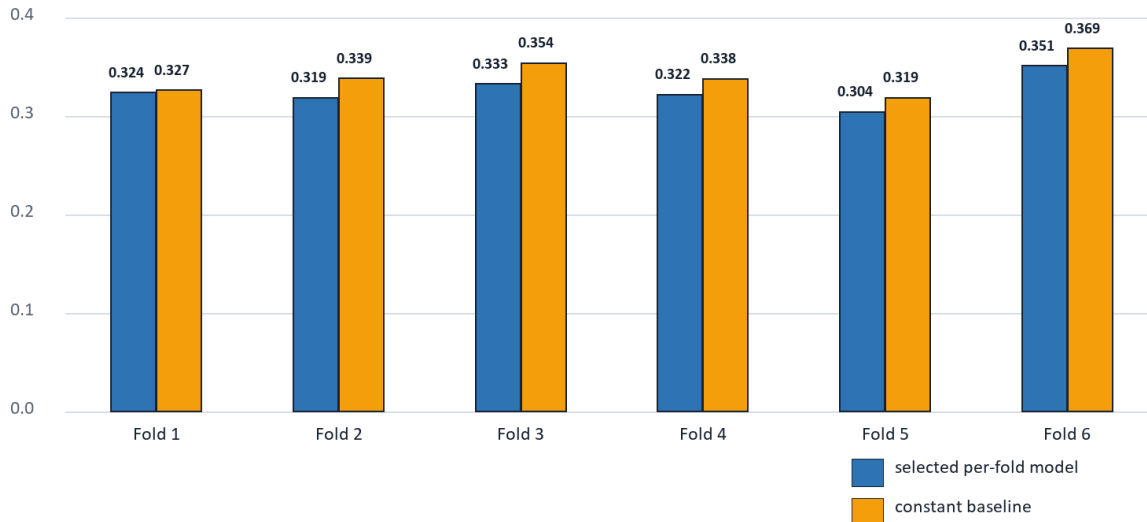


Figure 7. Conditional force MAE versus fold-fitted constant baseline

Source: v4 metrics.json [E27]. The selected family is chosen within each outer training partition, but the outer TEST groups are post-hoc.

Table 21. Force-estimation headline metrics and gates

Metric	Observed	Gate/benchmark	Status
Conditional MAE, retained isotonic	0.328 N	≤ 0.75 N	Numerically passes, but post-hoc
Displayed/end-to-end MAE, v2 replay	0.380 N	≤ 0.75 N	Numerically passes, but post-hoc
RMSE	0.384 N	≤ 1.0 N	Numerically passes, but post-hoc
Absolute force bias	0.115 N	≤ 0.20 N	Numerically passes, but post-hoc
Per-ROI conditional MAE	0.305-0.408 N	≤ 1.0 N	Numerically passes, but post-hoc
Cross-validated p95 absolute error	0.643 N	No frozen release gate	Uncertainty summary only
MAE gain vs. constant, retained	0.013 N / 3.81%	Combined resolution gate expected $\geq 5\%$ plus tracking	Fails material-improvement component
Expanded-search gain vs. constant	4.49%	$\geq 5\%$ target used in recovery audit	Fails
Mean Spearman rho	0.207 retained; 0.136 expanded	≥ 0.30 recovery criterion	Fails
Predicted/true SD ratio	0.207	≥ 0.25 recovery criterion	Fails
Force resolution	Not established	Required for resolution claim	Blocked
Common all-ROI safe range	Not established	Required by authoritative plan	Blocked

Source: [E02, E04, E27-E29].

Why the reported force MAE is not force resolution

Prediction spread is only 13-32% of true spread and rank correlation is weak across outer folds.

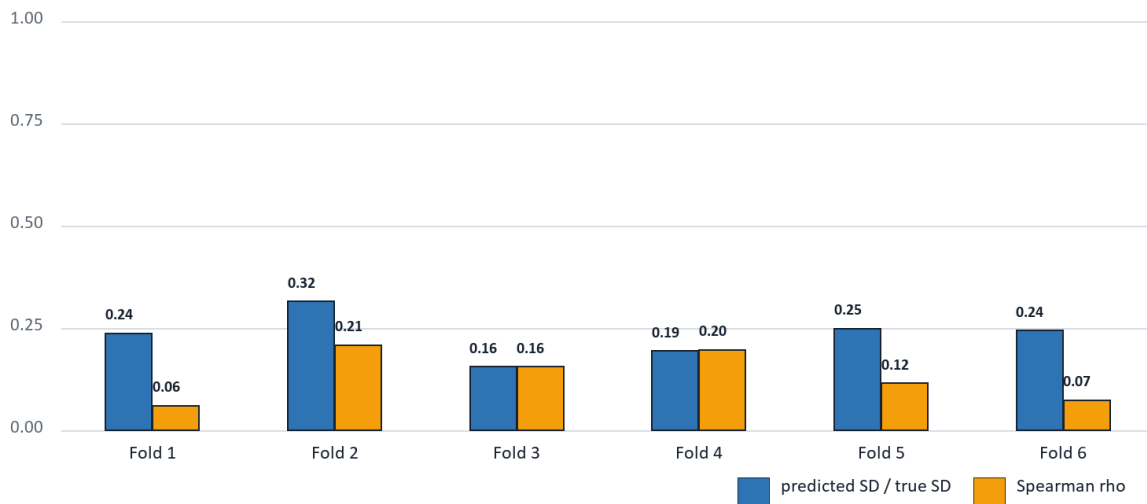


Figure 8. Fold-level force variation-tracking limitations

Source: v4 expanded candidate benchmark [E27]. Low spread and weak rank tracking show why MAE alone is insufficient.

11.3 Why low MAE does not establish force resolution

The evaluated force window is narrow and centered near roughly 2.3 N. A constant predictor can therefore achieve a mean MAE of 0.341 N without resolving meaningful within-session force changes. The isotonic model reduces that error only slightly and compresses predicted variation. Mean predicted standard deviation is about one-fifth of true standard deviation, and rank correlation is weak. In practical terms, the output often behaves like a mildly varying central estimate rather than a sensor that reliably orders or separates small force steps.

True resolution requires a dedicated randomized, settled small-step protocol with known increments, repeated across sessions and positions, plus a declared detection/discrimination rule. The collected manual presses and continuous automatic ramps do not provide that design. The NEF characterization metric is also only a noise-to-slope stability proxy and cannot substitute for a step-resolution experiment. [E03, E25-E27]

11.4 Release decision

- The six outer TEST folds were reused after inspection; no untouched confirmation data remain.
- The force estimator fails the combined improvement/rank/response-resolution gate.
- The authoritative all-ROI common safe-range procedure remains failed.
- Known-force physical validation of the live optical estimate is absent.
- End-to-end latency, 60-minute soak, and representative-user validation are absent.
- The model bundle and release_decision.json explicitly set validated_claim_allowed to false.

12. Historical Models and What They Contributed

Several earlier analyses remain in the workspace. They are useful for explaining technical evolution but are not directly comparable because they use different archives, force ranges, features, validation grains, and governance rules. Their numbers must not be blended into the v4 headline.

Table 22. Historical model record

Model/run	Data and validation	Headline result	Current interpretation
Automatic-archive linear	162 automatic sessions; grouped full sessions	Force all-frame MAE 0.526 N, R^2 0.0165; session localization 54.94%; non-contact predicted-contact rate 99.58%	Automatic archive is now forbidden for application modeling; diagnostic failure
Early manual ROI-conditioned linear	Older 9-session Manual Calibration.zip; within-session blocked fifths	Localization 79.68%; end-to-end force MAE 2.562 N, R^2 0.087	One session/ROI and frame-block validation; exploratory only
Early global manual families	Within-session validation; ridge/nonlinear/RFF	Best delayed nonlinear MAE 2.380 N, R^2 0.217	Diagnostic evidence that temporal context helped but generalization was weak
Clean global TEST2-TEST6	Leave-one-complete-TEST-out; TEST1/TEST7/no-contact excluded	Delayed nonlinear MAE 1.646 N, R^2 0.362; low-force MAE 2.391 N; false-contact rate 100%	Historical force-shape diagnostic; exclusions and low-force failure prevent deployment
Current characterization global summary	Session-zeroed total linear over 1.7-3.0 N	R^2 about 0.0136; constant often best median session MAE	Global aggregation sacrifices position information and does not support a universal force map
Recovery v1	Manual-only post-hoc six folds	Displayed MAE 0.479 N; localization macro F1 88.04%	First disclosed recovery; later runtime parity fixes supersede it
Recovery v2	Complete-sequence replay with debounce	Displayed MAE 0.380 N; forced localization F1 89.13%; force-resolution gate failed	Source of retained isotonic force mapping
Recovery v3	Event-centric camera-only output	Event localization F1 93.43%; no Newton output	Source of v4 detector/localizer
Hybrid v4	v3 event/localization + v2 isotonic force	Event F1 93.43%; conditional force MAE 0.328 N	Current experimental bundle; validated claim withheld

Source: [E04, E27, E31-E34] and historical analysis outputs. Metrics are intentionally not pooled.

12.1 Main lessons from the model history

- Frame-level or within-session splits can overstate generalization; independent session/TEST-group folds are necessary.
- Automatic cyclic data are not interchangeable with manual application presses; archive role and collection protocol matter.
- Total positive light is spatially heterogeneous and non-monotonic, so a single global force relationship is weak.
- Event aggregation and active-fraction spatial evidence are more reliable for discrete localization than global continuous force regression.
- A low error inside a narrow force band can coexist with weak dynamic tracking; constant-baseline and spread/rank checks are mandatory.
- Simple argmax/monotonic models can outperform or equal more complex families when the dataset is small and device-specific.

13. Physical, Software, and Engineering Validation

13.1 Physical camera and load-cell validation

Table 23. Measured physical acquisition results on 3 August 2026

Test	Measured result	Interpretation
DirectShow camera cadence	901 frames/60 s; 14.98485 FPS; interval median 64.105 ms, p95 79.980 ms, max 81.105 ms	Stable tested stream; nominal 30 FPS readback not authoritative
Nano/HX711 cadence	661 samples/60 s; median interval 91.847 ms (~10.89 Hz), p95 92.128 ms	No gaps or parse errors in tested run
200 g calibration windows	U=56,052.6±22.748 counts; L=178,197.388±25.410; factor=610.723941 counts/g; SNR=4,807; loaded CV=0.0208%	Calibration quality gates passed
Fixture preload and retare	Preload 54.216 g; retare 89,201.518±28.025 counts	Fixture-aware zero was necessary
Known-mass verification	199.0146±0.0450 g; error 0.9854 g / 0.4927%	Passed ±5% criterion on one placement
Accepted physical session	161 frame/video/feature/master rows; 125 load samples; 0 missing/invalid sync; peak 5.0935 N	Complete synchronized acquisition demonstrated
Physical-session sync gaps	Median 24.943 ms; p95 42.830 ms; max 45.940 ms	Well within 200 ms limit for this run
Physical optical response	No dominant ROI; peak mean delta-V below localization threshold	Session did not demonstrate specimen mechanoluminescence or live localization

Source: [E06].

This validation supports the tested acquisition/calibration path, not universal hardware performance. It did not measure true optical latency, destructive disconnect recovery, repeated placements, long-term drift, physical live-model accuracy, or specimen mechanoluminescence. The absence of a clear optical response in the accepted session is a negative result that must be preserved in the thesis.

13.2 Printer validation

Software implementation covered connection ownership, Marlin response parsing, command lifecycle, bounds, press-zero validity, sequence phases, pause/abort, force-limit polling, synchronized motion provenance, and simulation. Physical checks identified Marlin 1.1.6, read positions with M114, homed explicitly, and exercised small X/Y/Z moves. Operator inspection confirmed positive Z moves away from the specimen. A full physical repeated-press sequence, loaded force abort, worst-case abort latency, long-run motion reliability, and mechanical repeatability were not tested. [E07]

13.3 Engineering reviews

Five structured reviews covered architecture, camera service, load-cell/synchronization, data integrity, and GUI/usability. Findings were resolved to zero recorded blockers/majors/minors at the dated checkpoint. Important resulting controls include newest-only preview buffers, bounded analytical/recording queues, exact schema validation, atomic writer finalization, one serial owner, explicit readiness flags, immutable frame IDs, generation-based transaction safety, and fail-closed partial recovery. [E08]

13.4 Current automated verification

Table 24. Current test audit on 18 August 2026

Group	Passed	Scope
Core/model/data-quality	189	Live bundle, contracts, schemas, features, baseline, calibration, synchronization, characterization, splits
Exporter/firmware/graphs/logging/motion	37	Data and support services
Performance/reliability/controller/serial/simulation sources	91	Reliability and repeated-press logic
Camera/printer/Qt workers/recording/simulation smoke	83	Service and integration paths

Group	Passed	Scope
GUI-bearing files run individually	90	Camera settings/orientation, components, transactions, smoke, visualization, workflow, printer GUI
Total	490	All current test files passed in isolated groups

Source: Current local test execution; validation_checks.json.

`pip check` reports no broken requirements. A monolithic mixed Qt run reproduced a Windows access violation during pyqtgraph ROI garbage collection; the same files pass when run individually. A separate documented recording-finalization timing flake can also appear only in a monolithic run. These are test-harness/process-lifetime reliability issues that should be fixed, but they do not invalidate the isolated assertions. Conversely, isolated tests cannot substitute for the missing physical live-model, soak, and user validation.

14. Limitations and Threats to Validity

Central limitation: Localization and force do not have the same evidential status. The nine-class event localizer is the strongest model result, while the Newton-valued force output remains a narrow-range post-hoc approximation whose resolution, common safe range, and prospective physical accuracy have not been established.

14.1 Force-estimation limitations

Table 25. Force-estimation limitation register

Limitation	Why it matters	Required reporting/mitigation
No untouched confirmation set	All six complete TEST groups were inspected during recovery; cross-validation is no longer prospective.	Call all v4 metrics retrospective/post-hoc and collect a new frozen confirmation corpus.
Narrow force population	The evaluated window is labeled 1.7-3.0 N and is concentrated near its center; low MAE is easier for a near-constant predictor.	Always pair MAE with constant-baseline gain, rank correlation, prediction/truth spread, and support boundaries.
Force resolution not measured	Continuous presses do not create randomized, settled, known small increments; NEF is only a proxy.	Do not quote a minimum detectable force or resolution. Run a dedicated repeated step-discrimination study.
Weak variation tracking	Mean Spearman rho is 0.207 and predicted/true SD is 0.207; the estimate compresses real variation.	Describe the output as approximate event force, not a dynamically accurate force trace.
Marginal value over a constant	Retained improvement is 3.81%; expanded search is 4.49%, below the recovery target and unstable by selected family.	Do not use MAE alone to imply meaningful calibration or tracking.
No common all-ROI safe range	The authoritative preprocessing/range procedure failed because response support and monotonicity differ by ROI.	Do not claim one validated all-position transfer function.
Spatial heterogeneity and cross-talk	Sensitivity is zero/negative for several ROI; diagonal response share has a median of only 0.376 and off-target response can be large.	Keep the model device/layout-specific; investigate ROI-conditioned models and mechanics.
Noise, drift, hysteresis, and lag	No-contact drift, automatic loading/unloading separation, and variable system lag can change the optical score independently of force.	Rebaseline, control imaging/temperature/mechanics, and validate across time, speed, and loading direction.
Conditional support	The 32-knot model only returns a number between fitted optical-score knots; declared Newton labels are not identical to score support.	Withhold rather than extrapolate or clip; report `below_range`/`above_range` explicitly.
Single-contact model	The mapping cannot decompose contributions from simultaneous rods.	Withhold all force values in multi-ROI state until real multi-contact data and an identifiable model exist.
No live known-force validation	The accepted physical session verified synchronized acquisition but produced no clear optical localization response.	Run known-force live replay/fixture tests on the actual specimen before any deployment claim.
Device and geometry specificity	Preprocessing assumes one orientation, mirror, ROI layout, optical baseline, camera behavior, and tested mechanics.	Do not generalize to another camera, skin, mounting, ROI layout, or lighting without recalibration and new validation.

Source: Consolidated from [E02-E06, E25-E30].

The report intentionally places these limitations beside the force metrics and repeats them in the thesis-safe claims section. This prevents a reader from extracting the 0.328 N MAE without seeing that the model has weak within-session ordering, limited spread, post-hoc fold reuse, and no established physical resolution.

14.2 Construct validity

- The load cell measures fixture/reference force, while optical features measure camera V-channel changes inside fixed rectangles. Their association is not direct proof of a material constitutive relationship.
- The term localization means selection among nine predefined regions. It does not quantify continuous position error in millimetres, sub-ROI location, or contact-shape reconstruction.
- The system-lag estimator combines material, camera, USB, host, interpolation, and algorithm effects. It is not intrinsic mechanoluminescent response time.
- Automatic loading-unloading separation is labeled hysteresis, but it contains only two sessions per exact condition and may include drift, rate dependence, fixture effects, and synchronization error.
- Automatic hold duration and dwell summarize the available cycle protocol; they are not a designed long-duration creep experiment.
- Positive delta-V discards darkening for several summary features, while the v4 detector preserves signed spatial range. Results therefore depend on the chosen optical construct.

14.3 Internal and statistical validity

- Sessions, not frames, are the independent unit. Frame counts can describe coverage but must not be presented as independent sample size.
- Outer groups were complete TEST identities, which limits direct leakage, but repeated post-hoc inspection introduces researcher degrees of freedom and optimistic selection risk.
- Only 102 labeled single-press events support nine classes. Per-ROI estimates therefore have limited independent-event counts and should be accompanied by the fold/session design.
- Several metrics are medians of session-balanced bins or rows. A median cross-talk row need not sum to one, and aggregate hysteresis rows are not independent physical replications.
- Four automatic sessions contain one fewer usable cycle than expected; the validation artifact identifies them. Characterization passed its data-quality gate with that qualification.
- No formal confidence intervals or preregistered hypothesis tests were used for current model claims. The results are engineering performance estimates.

14.4 External validity and operational limitations

- Evidence is from one sensing assembly, tested camera path, one fixture/load-cell calibration, and a narrow set of laboratory conditions.
- Camera auto-exposure, focus, white balance, gain, ambient illumination, material aging, temperature, mounting, and mechanical boundary conditions can change optical response.
- The full Ender 3 repeated-press sequence and force-triggered abort were not physically validated; worst-case abort latency is unknown, and Marlin EMERGENCY_PARSER was disabled on the tested firmware.
- The 60-minute inference soak, measured camera-to-display p95 latency, memory-growth gate, reconnect/failure-injection campaign, and representative-operator study are incomplete.
- The accepted physical synchronized session did not demonstrate mechanoluminescence or a live localization result. Physical acquisition readiness is therefore stronger than sensor-phenomenon validation.
- Windows/Qt process-lifetime behavior prevents treating one monolithic test invocation as stable even though all 490 tests pass in isolated groups.

14.5 Data and software traceability limitations

Archive hashes, feature-store manifests, model hashes, exact schemas, and dated decision records provide substantial artifact traceability. However, the absence of committed Git history prevents authoritative code provenance, review attribution, and reproducible checkout by tag. The current local source may therefore be described as the 18 August 2026 workspace state, not as a versioned release. Future results should be generated from a committed, tagged, dependency-locked release with captured hardware/firmware identities.

15. Thesis-Safe Claims and Writing Guidance

15.1 Claim matrix

Table 26. What the thesis may and may not claim

Topic	Supported wording	Unsupported wording
Program	A standalone calibration/acquisition application was implemented with camera, load-cell, synchronization, recording, optional printer automation, and structured exports.	The complete system is deployment-certified or physically validated under all operating modes.
Data integrity	Two archives were immutably reconciled into 245 sessions and 179,176 decoded frames with zero missing/unmatched video frames.	All frames are independent samples or all archives are eligible for application modeling.
Load-cell reference	One 200 g calibration/verification on the tested fixture achieved 0.493% error and passed the project gates.	The load cell is universally calibrated across time, mounting, temperature, or force range.
Localization	In grouped retrospective evaluation, the camera-only event localizer achieved 93.43% macro F1 on 102 single-press events from 54 sessions.	The model is prospectively validated, continuously localizes position, or has measured physical multi-press accuracy.
Selective localization	Abstention produced 100% macro F1 at 82.93% coverage in the same post-hoc corpus.	The model is 100% accurate without stating coverage and post-hoc status.
Force	Inside fitted support, the experimental mapping had 0.328 N conditional MAE and 0.643 N p95 absolute error in retrospective grouped evaluation.	The sensor has 0.328 N resolution, measures force accurately in general, or supports extrapolation beyond the trained score/range.
Characterization	ROI-specific sensitivity, nonlinearity, noise/drift, repeatability, cross-talk, hysteresis, lag, and NEF-proxy metrics were computed under documented protocols.	The metrics establish intrinsic material constants, causal mechanisms, or broad environmental robustness.
Automation	Printer-control logic, safety states, simulation, protocol handling, and small manual moves were exercised.	A loaded repeated-press sequence and emergency force abort were physically validated.

Source: Claim synthesis governed by [E02-E08, E23-E30].

15.2 Copy-ready limitations paragraph

Thesis-ready text: Although the camera-only system produced strong retrospective event-localization performance, the force output must be interpreted as an experimental approximation rather than a validated force measurement. All six outer TEST groups had been inspected during post-hoc model recovery, leaving no untouched confirmation set. The retained isotonic mapping achieved a conditional MAE of 0.328 N within a narrow nominal 1.7-3.0 N band, but improved only marginally over a session-balanced constant, showed weak rank tracking, and compressed the observed force variation. A common all-ROI safe range and minimum resolvable force were not established, and no known-force physical validation of the live optical model was completed. The model therefore withholds force outside fitted optical-score support and during multi-contact states; its output should not be generalized to other devices, geometries, lighting conditions, or force ranges without prospective recalibration and validation.

15.3 Recommended results order

45. Establish the acquisition/calibration system and the immutable dataset before presenting model performance.
46. Report the failed original confirmatory gate before the post-hoc recovery, so the governance decision is transparent.
47. Present event detection and nine-class localization as the main successful modeling result.
48. Present force MAE beside constant-baseline gain, Spearman rho, spread ratio, p95 error, and support/withholding behavior.
49. Separate characterization findings by protocol: manual general metrics, automatic hysteresis/hold/dwell metrics, and physical hardware verification.
50. End with unresolved physical, prospective, soak, usability, multi-contact, and force-resolution validation rather than labeling the system fully validated.

15.4 Numbers that must retain their denominator

- 93.43% macro F1: 102 single-press events, 54 independent sessions, six complete TEST-group outer folds, post-hoc.
- 100% selective macro F1: only at 82.93% coverage and under the same post-hoc design.
- 0.328 N conditional MAE: only in-support contact frames in the narrow operating population; not missed-contact end-to-end performance or resolution.
- 0.493% verification error: one 200 g placement on one fixture after retare.
- 179,176 frames: decoded observations across 245 sessions, not 179,176 independent experimental replicates.
- Automatic hysteresis condition values: only two sessions per exact condition and descriptive aggregation.

16. Reproducibility and Transfer to ChatGPT Work

16.1 Artifact set delivered with this report

Table 27. Companion artifacts

Artifact	Purpose
SPARE_Calibration_Program_Full_Technical_Report_2026-08-18.docx	Primary presentable technical report with figures, tables, caveats, and appendices
SPARE_Calibration_Program_Full_Technical_Report_2026-08-18.md	Machine-readable full-text version for direct ingestion into ChatGPT Work
SPARE_Report_Validation_Companion.ipynb	Executable/embedded-output checks for headline data and model metrics
validation_checks.json	Machine-readable report QA and output hashes
evidence_manifest.csv	Evidence ID to workspace-path register
current_schema_dictionary.csv	Current authoritative 812-row schema/data dictionary
chart_map.csv	Chart question, takeaway, source, and filename map
charts/	Eight source-backed PNG figures used by the report

Source: Generated together by `build_report.py`.

16.2 Rebuild procedure

51. Open a PowerShell prompt at the workspace root and confirm that the ``spare_calibration_gui`` evidence tree is present.
52. Run ``python reports/thesis_technical_report_2026-08-18/build_report.py`` in an environment containing pandas, Pillow, and python-docx. The bundled Codex document runtime may also be used.

53. Review `validation\_checks.json`; every check must be true, the current test audit must remain explicitly qualified, and output hashes should be recorded.
54. Open the DOCX in Word and update the table of contents if required. Preserve the PDF/PNG visual-render QA record when the document is submitted.
55. When the underlying project changes, create a new dated report directory rather than overwriting this evidence cutoff.

16.3 Suggested ChatGPT Work ingestion prompt

Upload the DOCX and Markdown report together with `evidence\_manifest.csv`, `current\_schema\_dictionary.csv`, `validation\_checks.json`, and the validation notebook. Instruct ChatGPT Work to treat this report as the controlling technical evidence dossier; preserve all denominators and evidence-status labels; distinguish implementation, physical verification, characterization, post-hoc model performance, and missing validation; and never transform approximate force MAE into a resolution or general-accuracy claim. Ask it to cite evidence IDs in draft thesis sections so every assertion remains traceable to the local project artifact listed in Appendix D.

16.4 Reproducibility boundary

The report generator validates current metric files and schemas, but it does not re-extract 31 GB of compressed automatic video, retrain every historical model, or physically operate the hardware. Full end-to-end reproduction requires the two original archives, the exact extraction/alignment scripts, the recorded configuration and run manifests, the tested hardware/firmware, and a versioned software environment. Artifact-level reproducibility is currently stronger than historical code-revision reproducibility because Git history is absent.

17. Required Next Work and Priority Order

Table 28. Recommended validation and development backlog

Prior ity	Work package	Minimum completion evidence	Claim unlocked
P0	Version control and release freeze	Commit all project-owned sources/configs/tests; lock dependencies; tag model/report releases; preserve hashes	Auditable software provenance
P0	Prospective confirmation corpus	Predeclare protocol and gates; collect new manual sessions untouched by selection; retain dedicated no-contact sessions	Prospective detection/localization/force performance
P0	Known-force live optical validation	Randomized known loads/placements on the actual specimen with complete live replay and synchronized truth	Physical accuracy statement within a declared range
P0	Force-resolution experiment	Settled randomized small increments, repeated sessions/ROI, explicit discrimination rule and uncertainty	Minimum resolvable force—only if passed
P1	All-ROI operating range	Training-only support and monotonicity procedure repeated with adequate sessions/ROI	Common or ROI-specific validated force ranges
P1	Real simultaneous-contact study	Labeled two-or-more-rod combinations with identifiable reference forces	Measured multi-contact localization and possibly decomposed force
P1	Printer physical safety validation	Loaded repeated cycles, bounds/interlock tests, force abort and worst-case stop latency	Physical automated-press readiness
P1	Latency/soak/reconnect reliability	Measured camera-to-display p95; 60-minute soak; memory-growth and failure-injection results	Operational performance/reliability
P2	Environmental and longitudinal robustness	Lighting, camera settings, temperature, aging, remounting, day/operator and device variation	Defined generalization envelope
P2	Representative usability validation	At least five representative operators against frozen setup/recovery/state-recognition gates	Validated usability claim
P2	Qt monolithic-test stabilization	Eliminate pyqtgraph teardown access violation and recording-finalization timing flake	Stable all-in-one CI execution

Source: Priorities derived from [E02, E05-E08, E28].

18. Conclusion

The project has produced a substantive research platform: it can calibrate the load-cell reference, acquire and synchronize camera/load-cell/printer evidence, preserve raw and derived products, reconstruct an immutable 245-session feature store, and compute a broad sensor-characterization suite. Engineering controls—one device owner, monotonic host time, exact schemas, bounded queues, partial-file recovery, model hashing, and fail-closed output—make the software evidence more mature than a typical exploratory script collection.

The principal model contribution is event-centric camera-only localization among nine fixed regions. Its 93.43% retrospective macro F1 is scientifically useful when paired with the 102-event/54-session/six-fold denominator and permanent post-hoc disclosure. The project also demonstrates how selective abstention and independent rod gates can make uncertainty visible rather than forcing every output.

Force estimation remains the limiting claim. The present isotonic model offers a bounded approximate Newton output and defensible error summaries, but it neither tracks force variation strongly nor establishes resolution, a universal safe range, multi-contact force, prospective validity, or physical live accuracy. The thesis should therefore frame the project as a verified calibration/acquisition platform with promising discrete localization and an explicitly experimental force approximation. That framing is both technically accurate and stronger than overstating a force-sensor result the evidence does not yet support.

Appendix A. Metric and Formula Reference

Table 29. Core acquisition, preprocessing, and evaluation formulas

Quantity	Definition	Interpretation/caveat
Signed pixel delta	$d(x,y,t) = V(x,y,t) - \text{median_baseline}[V(x,y)]$	Preserves both brightening and darkening relative to the unloaded baseline
Positive pixel delta	$d+(x,y,t) = \max(d(x,y,t), 0)$	Ignores darkening; used by legacy light and active-area summaries
Integrated signed light	Sum of d over pixels in one ROI	Size-dependent unless divided by ROI area
Integrated positive light	Sum of $d+$ over pixels in one ROI	Cannot express negative optical response
Active fraction	$\text{count}[d > \text{threshold_V}] / \text{ROI pixel count}$; default threshold = 10 V	Dimensionless active-area estimate; threshold depends on camera/baseline stability
Centroid	First spatial moment of positive delta divided by total positive delta	Unavailable when positive mass is zero
Counts per gram	$c = (L - U) / M$	Signed calibration; L/U are loaded/unloaded mean counts and M is known mass in grams
Reference force	$F_gf = (\text{raw} - \text{tare}) / c$; $F_N = F_gf \times 0.00980665$	Direction is preserved by signed c ; gravity conversion is conventional
Online synchronized force	Exact timestamp; else bracket interpolation if both sides ≤ 200 ms; else nearest ≤ 200 ms; otherwise invalid	Never crosses device-session identity
Offline aligned force	Per-session linear interpolation at $\text{frame_time} - \text{fitted_lag}$	No extrapolation/cross-session interpolation; maximum bracket gap 300 ms
v4 contact score	peak-to-peak range of nine filtered normalized signed sums	Unitless, device/model-specific
Localization confidence	$(\text{largest accumulated evidence} - \text{second-largest}) / \text{total accumulated evidence}$	Used for selective abstention; not a calibrated probability
Conditional MAE	mean $ \text{estimated force} - \text{reference force} $ over eligible in-support contact observations	Excludes out-of-support/withheld cases; state the eligible population
Macro F1	Unweighted mean of class-specific F1 across nine ROI	Prevents large classes from dominating but remains sample-size sensitive
Hysteresis percentage	Loading-unloading separation divided by observed optical span for a matched force bin	Descriptive system response, not an intrinsic material constant
NEF proxy	No-contact optical noise divided by fitted low-force sensitivity	Stability proxy only; not force resolution

Source: [E13-E17, E25-E30].

Appendix B. Frozen Release-Gate Status

Table 30. Selected release gates and current disposition

Gate	Frozen criterion	Current evidence	Disposition
End-to-end force MAE	≤ 0.75 N	v2 displayed MAE 0.380 N	Numerical check passes post-hoc; release still blocked
Forced localization macro F1	≥ 0.80	v4 event macro F1 0.934	Numerical check passes post-hoc
No-contact frame FPR	≤ 0.05	maximum fold 0.0244	Numerical check passes post-hoc
Contact recall	≥ 0.90	minimum fold 0.942	Numerical check passes post-hoc
Per-ROI recall	≥ 0.70	minimum event recall 0.903	Numerical check passes post-hoc
Force resolution/tracking	Material improvement plus response tracking	Constant gain $< 5\%$; rho/spread checks fail	Fails
Common safe range	Training-only all-ROI supported interval	Authoritative procedure failed	Fails
Invalid/OOD exact output	0	Runtime fail-closed contracts and tests	Implemented/tested; physical fault campaign pending
Inference p95	≤ 30 ms	Performance tests exist	Software evidence only; verify on final deployment host
Camera-to-display p95	≤ 300 ms	Not measured end-to-end	Pending
Soak	60 min; memory growth ≤ 100 MB after warm-up	Not performed	Pending

Gate	Frozen criterion	Current evidence	Disposition
Representative usability	≥5 operators plus success/time/state gates	Not performed	Pending

Source: `config/live_sensor_release_gates.json` [E10] and [E04-E05, E27-E28].

Appendix C. Data and Artifact Inventory

Table 31. Primary data populations

Population	Sessions	Frames	Role	Integrity/status
Manual Calibration (2).zip	83	29,318	54 primary TEST sessions; 11 no-contact; 18 replay; only source for application modeling and general characterization	SHA-256 e039dcd49f3390d34dc979490dcf46a6ecc43d9e6a2aec08e4952acc14b5006a; 1,283,430,583 bytes
Auto Calibration (2).zip	162	149,858	Hysteresis and creep-related descriptive characterization only	SHA-256 f81a41b635aa2b893e9fa1faff0bda2db41ad0717d8c134fb6d907e1471e344d; 31,171,632,642 bytes
Immutable feature store	245	179,176	Exact decoded-video optical feature source for downstream analysis	0 missing; 0 unmatched; source archive hashes unchanged
v4 modeling subset	54	102 events	Six complete TEST-group outer folds	Manual primary sessions only; post-hoc outer-fold reuse
Dedicated no-contact	11	Frame replay	Warm-up/no-contact FPR evidence and noise/drift characterization	Manual archive; session-independent evidence
Physical accepted session	1	161 camera/master frames; 125 load samples	Acquisition/synchronization demonstration	0 missing/invalid synchronized rows; no clear optical localization response

Source: [E06, E23-E27]. Archive hashes/bytes are from immutable audit manifests.

Table 32. Current authoritative export schemas

Schema/product	Current columns/rows	Notes
frame_features.csv	366	Frame metadata, reference/sync, processing state, nine-ROI raw/derived/live-model fields
master.csv	385	Frame feature schema plus synchronized/provenance fields used by final trial product
loadcell_raw.csv	46	Raw serial identity/timing/counts, calibration/tare/force, integrity and motion provenance
All-export union	404	Union exposed by the current export/data-dictionary scope; sparse fields remain explicit
current_schema_dictionary.csv	812	One row per scoped output/field definition; 812 current rows
Historical schema checkpoint	647 then 734 rows	Dated intermediate counts preserved in earlier documentation; superseded by the current 812-row export

Source: `Current data/schemas.py` [E20] and generated `current_schema_dictionary.csv`.

Appendix D. Evidence Register

Evidence IDs used in the body map to current workspace-relative paths below. The CSV companion contains the same register for machine ingestion. Historical evidence is intentionally labeled so that it cannot silently override current decisions.

Table 33. Evidence ID to artifact map

ID	Workspace-relative path	Use/status
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ID	Workspace-relative path	Use/status
E01	spare_calibration_gui/README.md	Current program purpose, installation, UI, operating workflow, exports, and use restrictions; current implementation documentation
E02	spare_calibration_gui/LIVE_SENSOR_GUI_IMPLEMENTATION_PLAN.md	Authoritative manual-only application-model plan, gates, splits, and deployment contract; authoritative plan
E03	spare_calibration_gui/SENSOR_CHARACTERIZATION_MASTER_PLANNING.md	Manual-first characterization plan and permitted automatic-archive role; authoritative plan
E04	spare_calibration_gui/LIVE_SENSOR_DECISION_LOG.md	Dated model-governance and v1-v4 recovery decisions; decision record
E05	spare_calibration_gui/LIVE_SENSOR_IMPLEMENTATION_STATUS.md	Current implementation, metric, test, and release-gate status; status record
E06	spare_calibration_gui/PHYSICAL_HARDWARE_VALIDATION_20260803.md	Camera, load-cell, calibration, synchronization, and physical-session measurements; physical validation
E07	spare_calibration_gui/ENDER3_REPEATED_PRESS_IMPLEMENTATION_REPORT_20260804.md	Printer implementation, protocol tests, and remaining physical limitations; implementation validation
E08	spare_calibration_gui/ENGINEERING_REVIEWS.md	Architecture, acquisition, integrity, and GUI review findings; engineering review
E09	spare_calibration_gui/config/default_config.json	Runtime defaults for camera, processing, load cell, printer, recording, and display; configuration
E10	spare_calibration_gui/config/live_sensor_release_gates.json	Frozen acceptance gates; configuration
E11	spare_calibration_gui/config/manual_timestamp_alignment.json	Offline per-session timestamp-alignment contract; configuration
E12	spare_calibration_gui/config/manual_only_experimental_hybrid_v4.json	v4 hybrid model disclosure and operating contract; configuration
E13	spare_calibration_gui/processing/feature_extraction.py	HSV conversion and per-ROI optical feature formulas; source code
E14	spare_calibration_gui/processing/baseline.py	Baseline capture, provenance invalidation, and drift gate; source code
E15	spare_calibration_gui/processing/loadcell_calibration.py	Signed counts/g calibration, tare, verification, and unit conversion; source code
E16	spare_calibration_gui/processing/synchronization.py	Host-monotonic frame/load-cell synchronization; source code
E17	spare_calibration_gui/core/timestamp_alignment.py	Offline training-label timestamp interpolation; source code
E18	spare_calibration_gui/core/experimental_live_sensor.py	Hash-checked NumPy-only v4 runtime, state machine, event and withholding logic; source code
E19	spare_calibration_gui/core/live_sensor_contracts.py	Fixed layout/orientation and fail-closed scientific-state contract; source code
E20	spare_calibration_gui/data/schemas.py	Canonical CSV schemas and data-dictionary definitions; source code
E21	spare_calibration_gui/services/session_recorder.py	Incremental recording, partial artifacts, integrity validation, and finalization; source code
E22	spare_calibration_gui/arduino/hx711_nano_stream/hx711_nano_stream.ino	Nano/HX711 firmware and ASCII serial protocol; firmware
E23	spare_calibration_gui/analysis_outputs/live_sensor_study/feature_store/feature-store-c0ec762f1dc888a7/reconciliation_report.json	245-session/179,176-frame immutable feature-store reconciliation; derived evidence
E24	spare_calibration_gui/analysis_outputs/live_sensor_study/characterization/evidence-products-manual-first-authoritative/quality_report.json	Characterization data-quality checks; validated evidence
E25	spare_calibration_gui/analysis_outputs/live_sensor_study/characterization/evidence-products-manual-first-authoritative/validation_report.json	Characterization methodology validation and required caveats; validated evidence
E26	spare_calibration_gui/analysis_outputs/live_sensor_study/characterization/evidence-products-manual-first-authoritative/tables	Authoritative characterization tables; validated evidence
E27	spare_calibration_gui/models/live_sensor_experimental_hybrid_v4/metrics.json	v4 grouped retrospective model metrics; post-hoc experimental evidence
E28	spare_calibration_gui/models/live_sensor_experimental_hybrid_v4/release_decision.json	Explicit release blockers and prohibited validated claim; release decision
E29	spare_calibration_gui/models/live_sensor_experimental_hybrid_v4/force_model.json	32-knot monotonic isotonic approximation; experimental model bundle
E30	spare_calibration_gui/models/live_sensor_experimental_hybrid_v4/localization_model.json	Accumulated active-fraction argmax and selective confidence rule; experimental model bundle
E31	analysis_outputs/linear_models/validation_metrics.json	Historical automatic-archive linear model metrics; superseded exploratory evidence
E32	analysis_outputs/manual_linear_models/validation_metrics.json	Historical nine-session manual linear model metrics; superseded exploratory evidence
E33	analysis_outputs/manual_global_models_clean/validation_metrics.json	Historical TEST2-TEST6 clean global-force model metrics; superseded diagnostic evidence

ID	Workspace-relative path	Use/status
E34	reports/software_calibration_metrics_2026-08-17/validation_summary.json	Previous metrics-report QA and share-with-caveats classification; report QA
E35	spare_calibration_gui/tests	Current automated test suite; verification

Source: evidence_manifest.csv

Appendix E. Current v4 Bundle Details

Table 34. Current model contract

Component	Frozen setting
Bundle status	experimental; model_eligible_under_authoritative_plan=false; validated_claim_allowed=false
Camera inputs	Nine signed_delta_v_sum plus nine active_fraction features in exact row-major ROI order
Forbidden live/model inputs	Load-cell reference, printer state, automatic archive, or any substituted layout/feature order
Temporal preprocessing	Per-ROI robust normalization followed by causal nine-frame moving mean
Warm-up	At least 120 unloaded frames
Global contact threshold	max(bundle fallback 0.596388, warm-up q99 of contact score)
Independent rod threshold	max(4.0, warm-up q99.9 + 1.0); release ratio 0.60
Debounce	Two frames to acquire contact and two to clear
Localization	Event accumulation of positive normalized active-fraction evidence; forced argmax
Selective confidence	Withhold single-ROI label below 0.374959 unless independent rod gates identify active rods
Switching	Two-frame maintained single-ROI switch
Force input	Filtered normalized signed spatial range
Force mapping	32-knot nondecreasing isotonic piecewise-linear interpolation
Out-of-support force	Unavailable; no clipping or extrapolation
Multi-contact force	Withheld

Source: [E12, E18, E27-E30].

Table 35. Isotonic force-model knots

Knot	Optical score x	Approximate force y (N)
1	0.477605118	1.859181544
2	0.751592775	1.859181544
3	0.788626210	2.172439866
4	1.113573853	2.172439866
5	1.118188147	2.185426260
6	1.322150433	2.185426260
7	1.324638958	2.190428593
8	1.479506588	2.190428593
9	1.479735502	2.191505955
10	1.586792732	2.191505955
11	1.589544328	2.228555832
12	1.719042667	2.228555832
13	1.719859675	2.287808978
14	1.722302699	2.287808978
15	1.722606003	2.311346265
16	2.920311474	2.311346265
17	2.920849549	2.326517018
18	3.726645925	2.326517018
19	3.728657786	2.381354101
20	5.102421815	2.381354101
21	5.106250677	2.387292261
22	5.281037627	2.387292261
23	5.285886463	2.459550581
24	5.331901872	2.459550581
25	5.332365333	2.479518991
26	5.498393302	2.479518991
27	5.504041989	2.486511251
28	8.943242006	2.486511251
29	9.097742326	2.487490540

Knot	Optical score x	Approximate force y (N)
30	9.195497559	2.487490540
31	9.265416352	2.692665684
32	10.091376436	2.692665684

Source: *force_model.json* [E29]. Interpolation is allowed only within the first and last x knot.

Appendix F. ROI Geometry and Source Map

Table 36. Canonical nine-ROI layout after rotate-90°-clockwise and horizontal mirror

ROI	x	y	width	height
1	98	161	82	89
2	197	156	76	99
3	312	154	58	97
4	100	268	73	89
5	204	264	76	96
6	314	260	66	94
7	104	383	73	77
8	194	394	87	66
9	312	379	61	77

Source: Frozen layout hash/ID *roi-9fbc67c50ee3bc7145fa* [E02, E12, E19]. Raw 640×480 becomes processed 480×640.

Table 37. Key source-code map for methods writing

Method topic	Primary source
Application entry and CLI	<i>spare_calibration_gui/app.py</i>
Default device/processing configuration	<i>spare_calibration_gui/config/default_config.json</i>
Optical feature extraction	<i>spare_calibration_gui/processing/feature_extraction.py</i>
Baseline capture/drift	<i>spare_calibration_gui/processing/baseline.py</i>
Load-cell calibration	<i>spare_calibration_gui/processing/loadcell_calibration.py</i>
Online synchronization	<i>spare_calibration_gui/processing/synchronization.py</i>
Offline timestamp alignment	<i>spare_calibration_gui/core/timestamp_alignment.py</i>
v4 live inference	<i>spare_calibration_gui/core/experimental_live_sensor.py</i>
Scientific-state/layout contracts	<i>spare_calibration_gui/core/live_sensor_contracts.py</i>
Schemas/data dictionary	<i>spare_calibration_gui/data/schemas.py</i>
Incremental recorder/finalization	<i>spare_calibration_gui/services/session_recorder.py</i>
HX711 firmware	<i>spare_calibration_gui/arduino/hx711_nano_stream/hx711_nano_stream.ino</i>
Model bundle	<i>spare_calibration_gui/models/live_sensor_experimental_hybrid_v4/</i>
Characterization tables	<i>spare_calibration_gui/analysis_outputs/live_sensor_study/characterization/evidence-products-manual-first-authoritative/tables/</i>
Current tests	<i>spare_calibration_gui/tests/</i>

Source: Evidence map [E09-E22, E26, E35].